

# Hidden keys and repeated biometric records

Can multiple protected records reveal identity when keys remain hidden?

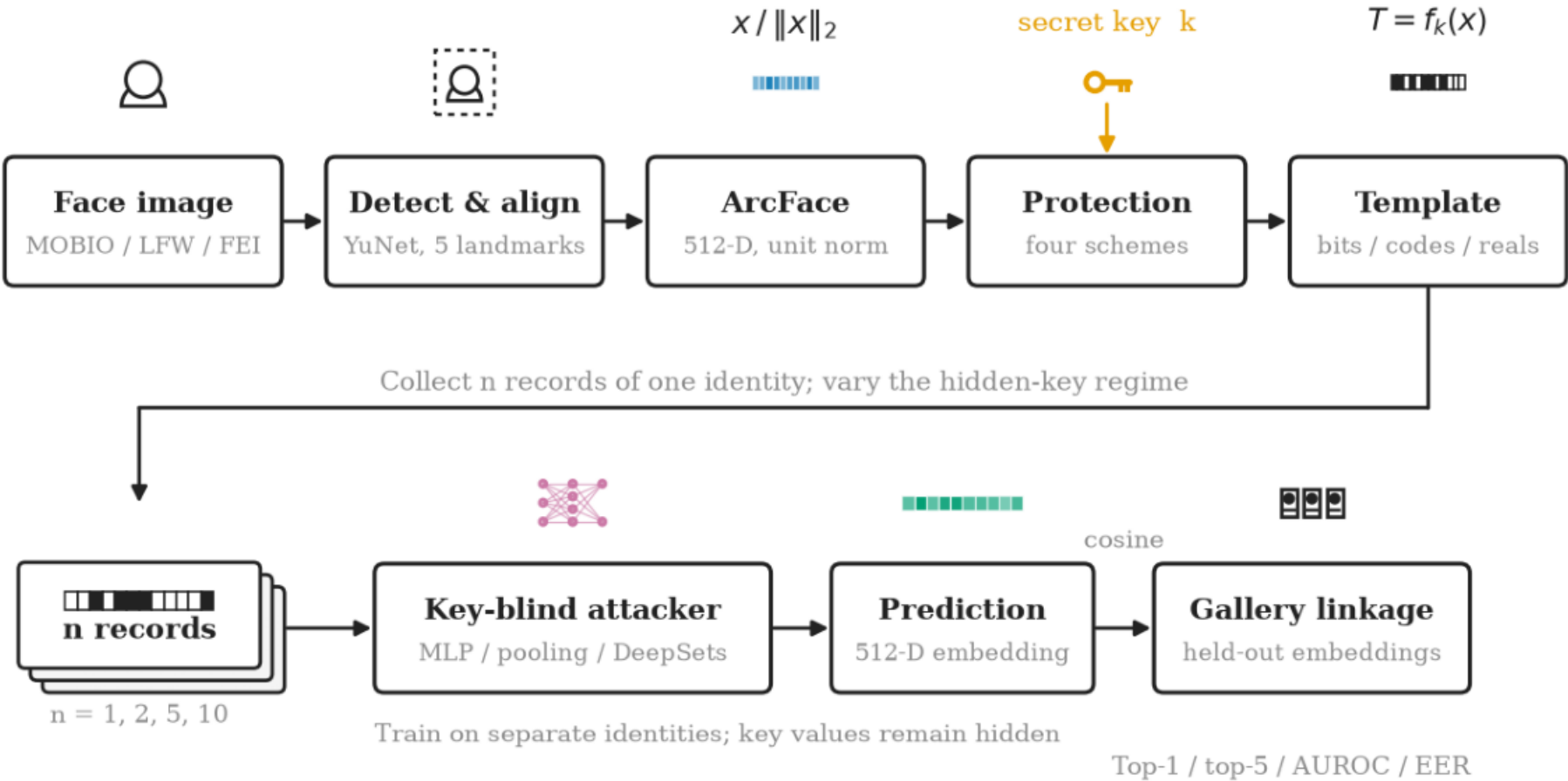
Fresh-key learned attacks: uncertainty remains; no equivalence claim.

Recurring transforms: large gains from multiple same-identity records.

Boundary location changes with the dataset and identity partition.

Independent study. No exact benchmark reproduction or universal privacy claim.

# Experimental architecture



Train on separate identities. Reconstruct an embedding, then link to a held-out gallery.

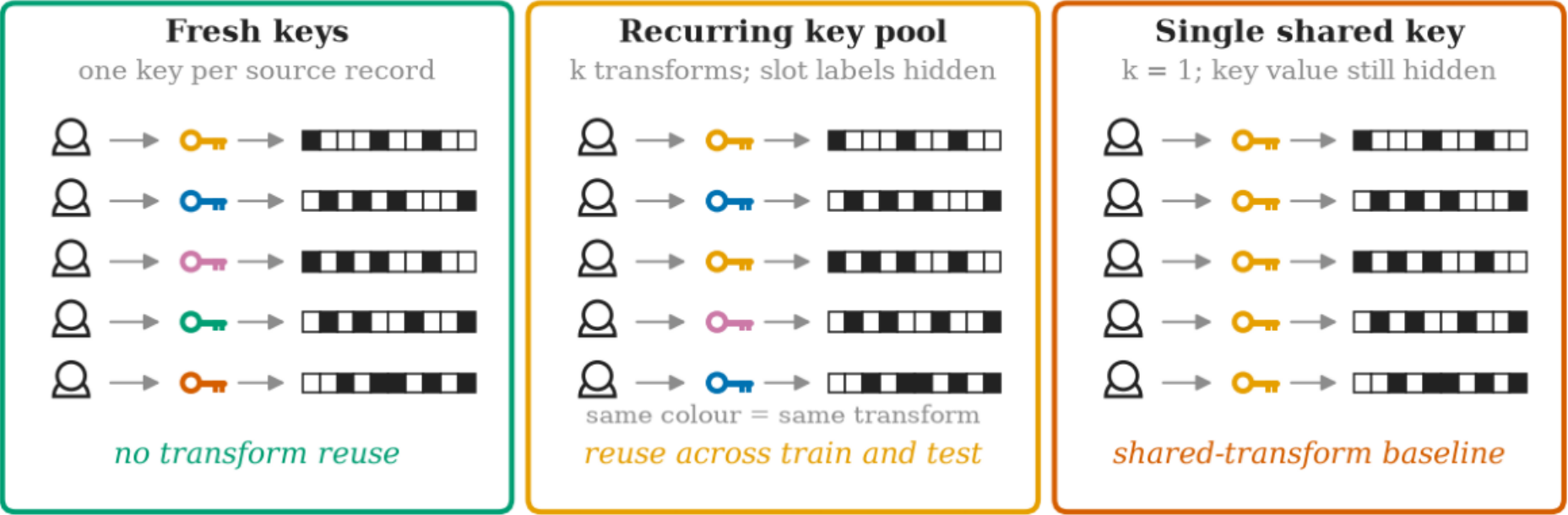
# Dataset and protection coverage

| Dataset | Identities | Train/val/test | Embeddings    | Variation       |
|---------|------------|----------------|---------------|-----------------|
| MOBIO   | 150        | 90/30/30       | 1,799 / 1,800 | Sessions        |
| LFW     | 125        | 75/25/25       | 1,500 / 1,500 | In-the-wild     |
| FEI     | 200        | 120/40/40      | 2,378 / 2,400 | Pose/expression |

BioHash: 128 bits; three datasets; includes a Haar-sign control on MOBIO.  
MLP-Hash: 512 bits; MOBIO; paper-specified, not source-exact.  
IoM-GRP: 300 categorical codes (q=16); MOBIO/FEI pilots.  
PolyProtect: 170 real values (m=5, overlap=2); MOBIO/FEI pilots.

FEI: 22 low-illumination failures; all identities retain a gallery image plus 10 exposures.

# What changes between key regimes?

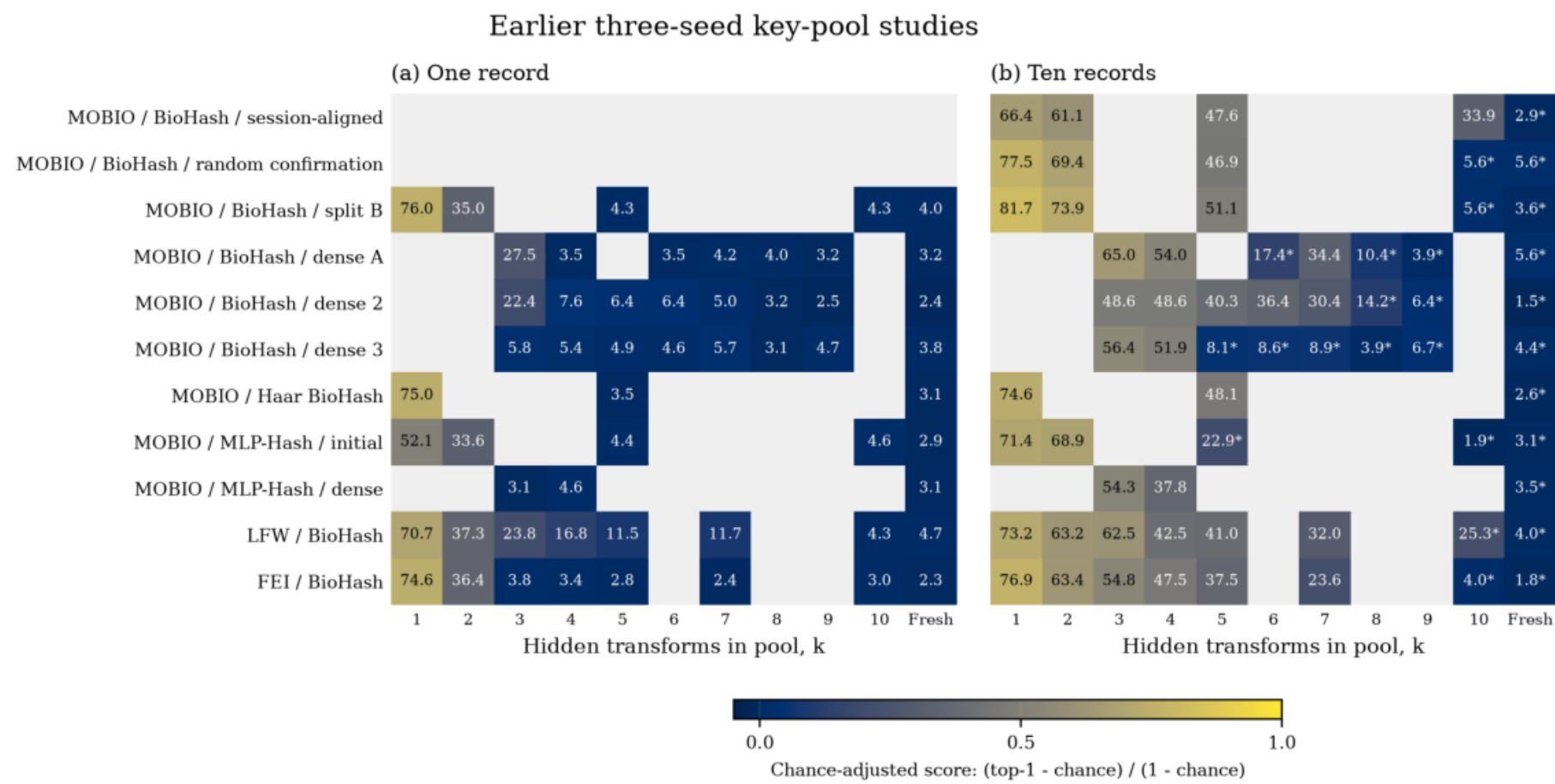


**Learned attacks inconclusive**   **Multi-record gains observed**   **Strong linkage observed**  
PolyProtect outside the theorem   boundary depends on dataset and split   hidden does not mean unlearnable

**Protocol**   Disjoint train / validation / test identities; one held-out gallery image per identity  
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals

Key values stay hidden in every regime. Recurring transforms are shared across identity splits.

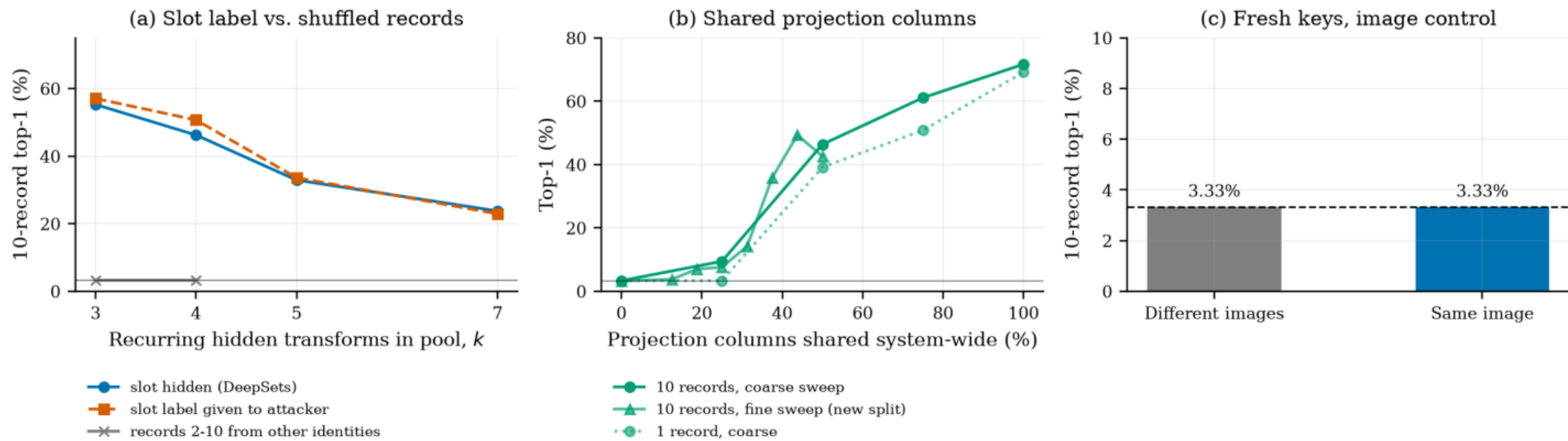
# Earlier three-seed key-pool studies



65 conditions from 11 earlier studies. New one-seed pilots are reported separately.

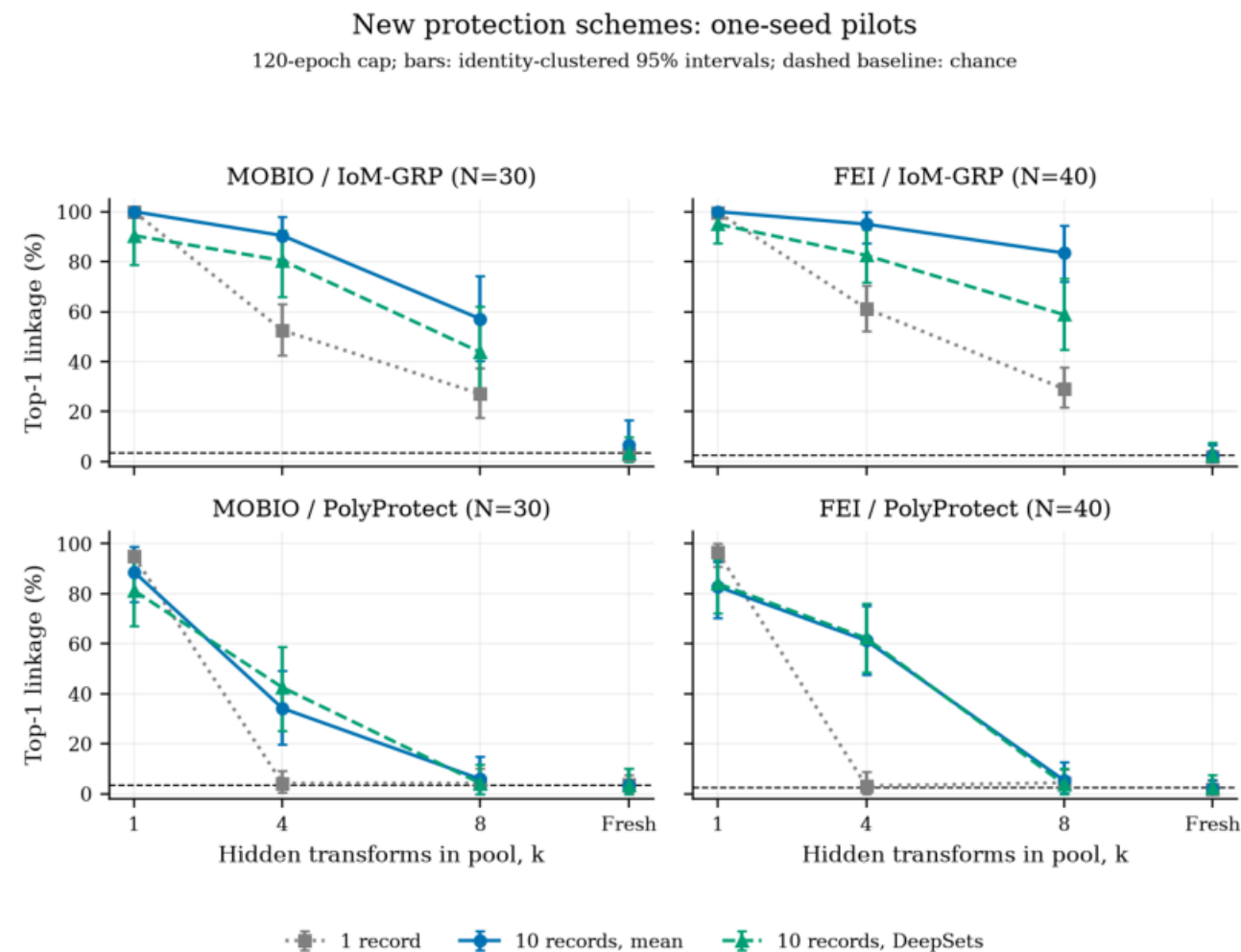
# Mechanism and correlation controls

MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Slot identifiers are not key values. Coarse and fine correlation sweeps use separate partitions.

# Approved scheme pilots: one seed



Fresh PolyProtect native matching: MOBIO 12.73%, FEI 13.73%; separate task, not a privacy result.

# Ready to draft; not yet ready to submit

Completed: two new schemes, 16 pilot cells, paired intervals, per-seed results.

Before submission:

1. Obtain authorized access to the next dataset.
2. Authorize staged confirmation; one-seed pilots are not confirmation.
3. Approve statistical margins and a multiple-comparison plan.
4. Independently review the corrected theorem and related work.
5. Obtain Sani's scientific and presentation review.

A/A\* is a venue ambition, not an established property or an acceptance prediction.